

EXPERIMENT:11 BITWISE OPERATOR

Q1. Write a program to apply bitwise OR, AND and NOT operators on bit level.

Algorithm:

Step 1: Start Step 2: Declare two integers a and b Step 3: Read input values for a and b Step 4: Compute

- $AND = a \& b$
- $OR = a | b$
- $NOT = \sim a$ (bitwise NOT of a)

Step 5: Display results Step 6: Stop

Code:

```
#include <stdio.h>

int main() {
    int a, b;
    int and_result, or_result, not_result;

    printf("Enter value of a: ");
    scanf("%d", &a);

    printf("Enter value of b: ");
    scanf("%d", &b);

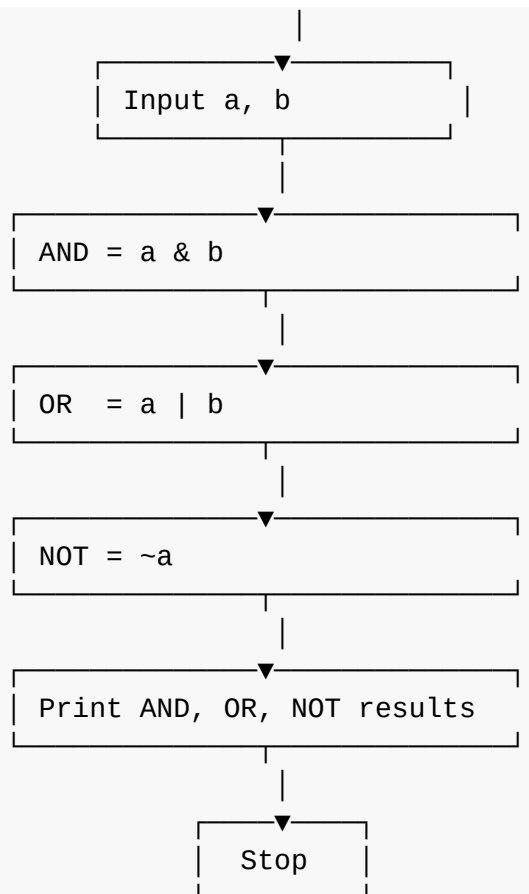
    // Bitwise operations
    and_result = a & b;    // AND
    or_result  = a | b;    // OR
    not_result = ~a;       // NOT (only on a)

    printf("\n---- Bitwise Results ----\n");
    printf("a & b (AND) = %d\n", and_result);
    printf("a | b (OR)  = %d\n", or_result);
    printf("~a (NOT)   = %d\n", not_result);

    return 0;
}
```

Flowchart:





Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
cd "/home/aaditya/experiments-in-c/exp11/" && gcc exp11Q1.c -o exp11Q1 && "/home/aaditya/experiments-in-c/exp11/"exp11Q1
aaditya@fedora:~/experiments-in-c$ cd "/home/aaditya/experiments-in-c/exp11/" && gcc exp11Q1.c -o exp11Q1 && "/home/aaditya/experiments-in-c/exp11/"exp11Q1
Enter value of a: 1
Enter value of b: 2

---- Bitwise Results ----
a & b (AND) = 0
a | b (OR) = 3
~a (NOT) = -2
aaditya@fedora:~/experiments-in-c/exp11$
```

Q2.write a program to apply left shift and right shift operator.

Algorithm:

Step 1: Start Step 2: Declare an integer variable num and shift amount shift Step 3: Read num and shift from the user Step 4: Compute

- left = num << shift
- right = num >> shift

Step 5: Display results Step 6: Stop

Code:

```
#include <stdio.h>

int main() {
    int num, shift;
    int left_result, right_result;

    printf("Enter a number: ");
    scanf("%d", &num);

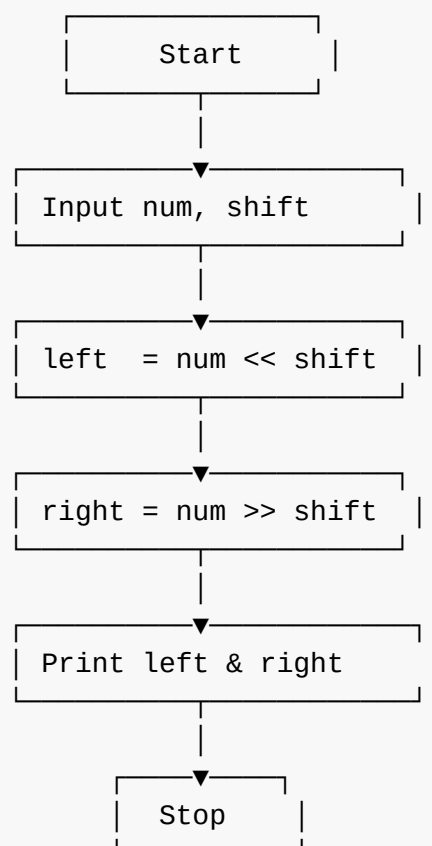
    printf("Enter shift amount: ");
    scanf("%d", &shift);

    // Perform shift operations
    left_result = num << shift;    // Left shift
    right_result = num >> shift;   // Right shift

    printf("\n--- Shift Results ---\n");
    printf("Left Shift (num << shift) = %d\n", left_result);
    printf("Right Shift (num >> shift) = %d\n", right_result);

    return 0;
}
```

Flowchart:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
cd "/home/aaditya/experiments-in-c/exp11/" && gcc exp11Q2.c -o exp11Q2 && "/home/aaditya/experiments-in-c/exp11/"exp11Q2
aaditya@fedora:~/experiments-in-c$ cd "/home/aaditya/experiments-in-c/exp11/" && gcc exp11Q2.c -o exp11Q2 && "/home/aaditya/experiments-in-c/exp11/"exp11Q2
Enter a number: 12
Enter shift amount: 5

--- Shift Results ---
Left Shift (num << shift) = 384
Right Shift (num >> shift) = 0
aaditya@fedora:~/experiments-in-c/exp11$
```